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# The Semiconductor Fab Handbook

A Bottom-Up Guide to Building, Operating, and Profiting  
From a 28nm/45nm Semiconductor Fab

# The Semiconductor Fab Handbook

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*White paper for new semiconductor manufacturers, investors, materials suppliers, and  
ecosystem planners*

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## Executive Summary

Semiconductors have become one of the world's most strategically important industries, underpinning artificial intelligence, cloud computing, automotive electronics, industrial automation, communications infrastructure, and consumer devices. As governments and corporations invest billions of dollars into new manufacturing capacity, a fundamental question remains:

### What does it take to build and operate a successful semiconductor fabrication facility?

Building a semiconductor fab is not simply a construction project. It is a long-cycle manufacturing capability project that combines process technology, equipment, materials, utilities, workforce development, yield learning, customer qualification, supply-chain resilience, and financial discipline.

For a new manufacturer, the central question is not only whether a fab can be built, but whether it can reach:

- Stable yield
- Qualified customers
- High utilization
- Reliable utilities
- Controlled materials
- Competitive wafer cost
- Sustainable profitability

A full-scale 300mm mature-node fab operating at 40,000-50,000 wafer starts per month (WSPM) requires significant capital, technology access, operational expertise, and customer demand. As a public benchmark, Tata Electronics' Dholera 300mm fab has been described as a 50,000 wafers-per-month facility spanning 28nm to 110nm technologies, with an announced investment of approximately US\$11 billion.<sup>1-3</sup>

For most new entrants, a pure 28nm-first strategy is rarely the optimal path. A more realistic approach is to establish a 300mm **mature-node** platform, initially ramping more mature (90-45nm technologies before introducing 28nm. This progression allows the manufacturer to build stable process control, low defectivity, customer-qualified process design kits (PDKs), reliable utility infrastructure, and predictable yield performance.<sup>4</sup>

This white paper examines what it takes to build and operate a semiconductor manufacturing fab from the ground up. Through a structured end-to-end assessment (using a representative base case as below), we analyze the technology, infrastructure, equipment, materials, utilities, workforce, yield, supply-chain, customer, and financial requirements needed to transform a greenfield fab investment into a sustainable and profitable semiconductor manufacturing business.

## Representative Base-Case Fab used for this study:

Reasons for these assumptions are explained in the next sections.

| Metric                   | Base-Case Assumption              |
|--------------------------|-----------------------------------|
| Wafer size               | 300mm                             |
| Installed capacity       | 40,000 WSPM                       |
| Target utilization       | 85%                               |
| Effective wafers/year    | 408,000 wafers                    |
| Technology path          | 45/40nm ramp → 28nm expansion     |
| Cleanroom area           | 20,000-40,000 m <sup>2</sup>      |
| Employees                | ~2,500-3,000                      |
| CAPEX range              | US\$7B-10B (planning estimate)    |
| Revenue potential        | ~US\$1.0B/year at US\$2,500/wafer |
| Yield target             | >85% mature functional yield      |
| Tool availability target | >90% for critical tools           |

### **Key takeaway:**

*A fab becomes successful only when the building, tools, process, materials, people, customers, and economics all work together. A cleanroom without qualified customers is a depreciation machine. A process without yield is not a business. A 28nm fab without high utilization is financially fragile.*

## 1. The Technology-Market Decision

The first decision is not technical; it is commercial. Before selecting a process technology, a new manufacturer must determine which node offers the optimal balance between market demand, manufacturing complexity, capital intensity, customer requirements, and long-term competitive differentiation.

This section evaluates the relative attractiveness of key semiconductor nodes and identifies the factors that influence node selection for a new fab entrant, including process complexity, capital expenditure requirements, ecosystem maturity, customer demand, and long-term market potential.

### 1.1 Technology Node comparison

When discussing semiconductor manufacturing, attention is often focused on leading-edge technologies such as 3nm, 2nm, and future process nodes. However, the majority of semiconductor volume continues to be produced on mature and mid-mature technologies. Automotive electronics, industrial automation, communication infrastructure, consumer devices, display controllers, power management ICs, and many IoT products do not require the most advanced transistor geometries to remain competitive.

| Node        | Typical applications                                           | Complexity  | Capex  | Market attractiveness |
|-------------|----------------------------------------------------------------|-------------|--------|-----------------------|
| 180nm       | Automotive, industrial, analog, power                          | Low         | Low    | Mature                |
| 130nm       | Microcontroller unit (MCU), analog, automotive                 | Low-medium  | Low    | Mature                |
| 90nm        | MCU, connectivity, industrial                                  | Medium      | Medium | Mature                |
| 65nm        | MCU, connectivity, display, embedded logic                     | Medium      | Medium | Strong                |
| 45/40nm     | MCU, automotive, connectivity, display, mixed-signal           | Medium-High | High   | Strong                |
| 28nm        | Connectivity, display, automotive logic, edge AI, consumer SoC | High        | High   | Very strong           |
| 22nm FD-SOI | Automotive, IoT, RF, low-power logic                           | High        | High   | Niche/specialty       |

For a new semiconductor manufacturer, the critical question is therefore not: "What is the most advanced node available?" but rather: **"Which node provides the best balance between market demand, manufacturing complexity, capital investment, and long-term profitability?"**

Among mature and mid-mature technologies, **28nm** has emerged as one of the industry's most commercially successful process nodes, often referred to as the industry's "golden node", the reasons for which are discussed in the next section.

## 1.2 Why 28nm Became a Strategic "Golden Node"

28nm technology node occupies an unusual position in semiconductor manufacturing, as it is in between traditional mature technologies and advanced nodes. This node provides substantial improvements in transistor density, power efficiency, and integration capability while avoiding the extreme complexity and cost associated with FinFET and EUV-based manufacturing.<sup>5</sup>

The commercial success of 28nm can largely be attributed to six structural advantages:

- Strong balance between performance and manufacturing cost
- Broad applicability across multiple end markets
- Long commercial lifecycle
- Attractive return on invested capital
- Large and mature design ecosystem
- Extensive global manufacturing infrastructure

Unlike older mature nodes, 28nm delivers sufficient performance for complex digital, connectivity, display, and automotive applications. At the same time, it avoids many of the manufacturing challenges introduced by FinFET-based technologies.

| Feature                  | 28nm                                    | 20nm                          | 16nm / 14nm and below                  |
|--------------------------|-----------------------------------------|-------------------------------|----------------------------------------|
| Transistor type          | Mostly planar CMOS, often HKMG variants | Planar, more complex          | FinFET                                 |
| Lithography              | DUV immersion / dry DUV                 | DUV multiple-patterning heavy | DUV + EUV or advanced multi-patterning |
| Manufacturing complexity | High but manageable                     | Higher                        | Very high                              |
| Design cost              | High                                    | Higher                        | Very high                              |
| Economic position        | Strong mature-node sweet spot           | Transitional                  | Advanced-node economics                |

From a manufacturing perspective, 28nm is particularly attractive because it remains a predominantly planar CMOS technology, allowing fabs to leverage mature **DUV lithography** infrastructure while achieving significantly higher integration levels than 45nm or 65nm processes.

## 1.3 Why choose even more mature nodes (e.g. 45/40nm) as a new manufacturer?

### Advantages of 28nm

A manufacturer may choose 28nm because it offers:

- Higher transistor density
- Lower cost per function
- Better power-performance characteristics
- Smaller die size for complex designs
- Strong relevance for:
  - Connectivity ICs
  - Display controllers
  - Advanced MCUs
  - Edge AI processors
  - Consumer SoCs
  - Automotive control systems
- Higher potential wafer average selling price (ASP) compared with older mature nodes. Public estimates place 28nm 300mm wafer pricing around US\$ 2,550 – US\$ 3,450, with an average near US\$ 3,000.<sup>7</sup>

The node continues to support high-volume production across consumer, industrial, networking, and automotive markets while remaining accessible to DUV-based manufacturing.

### **Advantages of 45/40nm**

However, a new manufacturer may instead prioritize 45/40nm because it provides:

- Lower process complexity.
- Faster yield learning.
- Lower defect sensitivity.
- Simpler lithography requirements.
- Lower customer qualification risk.
- Strong automotive and industrial adoption.
- Good wafer ASP in the estimated range of US\$1,800 – 2,500 per 300mm wafer.

For a first-time fab operator, these factors can significantly reduce execution risk during the early years of operation.

### **Strategic Recommendation**

For most new semiconductor manufacturers, the optimal strategy is not to begin with high-volume 28nm production immediately. A more practical roadmap is:

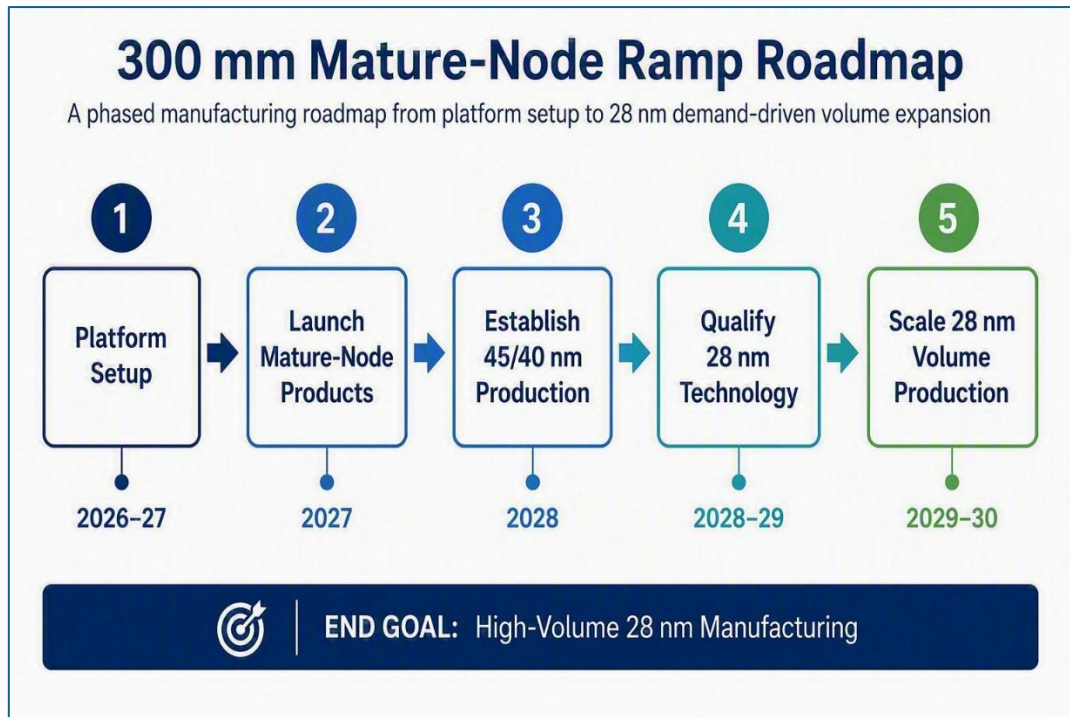


Figure 1: Simplified roadmap illustrating the phased evolution of a 300 mm fab from platform setup to high-volume 28 nm manufacturing.

This phased progression allows the organization to build the following capabilities before introducing the additional complexity associated with large-scale 28nm production:

- Process integration expertise.
- Yield-management capability.
- Metrology and inspection discipline.
- Manufacturing culture.
- Customer confidence.
- Supply-chain maturity.

#### **Key takeaway:**

*Mature nodes remain the foundation of modern semiconductor manufacturing. Among them, 28nm represents one of the best balances ever achieved between performance, manufacturing complexity, ecosystem maturity, capital efficiency, and market demand, making it a highly attractive destination for new semiconductor manufacturers. However, a new manufacturer should usually build a 45/40nm-capable platform first and use it as the operational bridge to 28nm.*

## 2. The Hypothetical Fab Definition

Before evaluating equipment, materials, workforce, or financial performance, the manufacturing strategy must be clearly defined. The selected technology node, capacity, product mix, and business model determine virtually every aspect of fab design, including:

- Cleanroom size
- Equipment requirements
- Utility demand
- Workforce requirements
- Process complexity
- Supply-chain requirements
- Capital expenditure
- Customer qualification requirements

For this white paper, a representative mature-node foundry is modelled to evaluate the technical and economic feasibility of a new semiconductor manufacturing operation.

**Important:** This is an analytical base case. Actual cost and ramp timing depend heavily on country, subsidies, labour cost, utility reliability, local ecosystem, equipment availability, technology partner, process complexity, and construction strategy.

### 2.1 Base-case fab specification

#### Commercial Definition

| Parameter           | Base-Case Assumption                                               |
|---------------------|--------------------------------------------------------------------|
| Fab Type            | Greenfield 300mm Foundry                                           |
| Business Model      | Pure-play foundry                                                  |
| Technology Platform | 45/40nm initial production, 28nm expansion                         |
| Wafer Size          | 300mm                                                              |
| Installed Capacity  | 40,000 WSPM                                                        |
| Utilization Target  | 85-90%                                                             |
| Ramp Period         | 5 years                                                            |
| Primary Markets     | Automotive, Industrial, Connectivity, Display, MCU, Edge Computing |
| Packaging Strategy  | External OSAT during initial ramp                                  |
| Technology Source   | Licensed platform or technology-transfer partner                   |

## Technology Platform Assumptions (45/40nm Initial Production with 28nm Expansion)

| Parameter                        | Base-Case Assumption                                          |
|----------------------------------|---------------------------------------------------------------|
| Technology Platform              | 45/40nm initial production, expandable to 28nm                |
| Transistor Architecture          | Planar CMOS                                                   |
| Gate Stack                       | Poly-Si gate for 45/40nm; High-K Metal Gate (HKMG) for 28nm   |
| Isolation Scheme                 | Shallow Trench Isolation (STI)                                |
| Source/Drain Formation           | Ion implantation with advanced anneal integration             |
| FEOL Complexity                  | High (45/40nm), Very High (28nm)                              |
| FEOL Process Steps               | ~200-300 (45/40nm), ~250-350 (28nm)                           |
| FEOL Mask Layers                 | ~15-20 (45/40nm), ~18-25 (28nm)                               |
| Critical FEOL Lithography Layers | ~8-12 (45/40nm), ~12-18 (28nm)                                |
| MOL Complexity                   | Contact and local interconnect formation                      |
| MOL Process Share                | ~10-15% of total process flow                                 |
| BEOL Metal Layers                | 8-10 (45/40nm), 10-12 (28nm)                                  |
| BEOL Process Share               | ~35-45% of total process flow                                 |
| Lithography Platform             | ArF Immersion (193nm), ArF Dry (193nm), KrF (248nm)           |
| Lithography Burden               | High (45/40nm), Very High (28nm)                              |
| Metrology Burden                 | High (45/40nm), Very High (28nm)                              |
| Total Mask Count                 | ~35-45 (45/40nm), ~40-55 (28nm)                               |
| Total Process Steps              | ~450-650 (45/40nm), ~600-800 (28nm)                           |
| Fabrication Cycle Time           | ~50-80 days (45/40nm), ~60-90 days (28nm)                     |
| Design Enablement                | Standard-cell libraries, SRAM compilers, I/O libraries        |
| Process Design Kit (PDK)         | Full customer-qualified PDK                                   |
| Reliability Capability           | Automotive and industrial qualification support               |
| Defect Density Target            | <0.1 defects/cm <sup>2</sup> at mature production             |
| Target Functional Yield          | 85-95% at mature production                                   |
| Process Integration Difficulty   | High (45/40nm), Very High (28nm)                              |
| Technology Source                | Licensed platform, JV partner, or technology-transfer partner |

## Physical Scale Assumptions

| Parameter           | Planning Value               |
|---------------------|------------------------------|
| Total site area     | 100-250 acres                |
| Fab building area   | 40,000-80,000 m <sup>2</sup> |
| Cleanroom area      | 20,000-40,000 m <sup>2</sup> |
| Utility area        | 15,000-30,000 m <sup>2</sup> |
| Warehouse area      | 5,000-15,000 m <sup>2</sup>  |
| Office/support area | 5,000-10,000 m <sup>2</sup>  |

## Process Complexity Assumptions (Compiled)

| Metric                         | 45/40nm    | 28nm       |
|--------------------------------|------------|------------|
| FEOL Masks                     | 15-20      | 18-25      |
| MOL Masks                      | 2-5        | 3-5        |
| BEOL Masks                     | 20-25      | 22-30      |
| Total Masks                    | 35-45      | 40-55      |
| FEOL Steps                     | 200-300    | 250-350    |
| MOL Steps                      | 50-80      | 70-100     |
| BEOL Steps                     | 200-270    | 280-350    |
| Total Steps                    | 450-650    | 600-800    |
| Critical Lithography Layers    | 15-20      | 20-30      |
| Fabrication Cycle Time         | 50-80 days | 60-90 days |
| Lithography Burden             | High       | Very High  |
| Metrology Burden               | High       | Very High  |
| Process Integration Difficulty | High       | Very High  |

*Important: These are planning-level assumptions. Actual numbers depend on process architecture, product type, foundry platform, reliability options, and customer requirements.*

## Why This Configuration?

This configuration was selected because it represents the most realistic entry point for a new foundry operator:

- Large enough to achieve manufacturing scale
- Mature enough to reduce technology risk

- Advanced enough to attract connectivity, automotive, and display customers
- Does not require EUV infrastructure
- Supports long-term expansion into higher-value nodes

## 2.2 Manufacturing Targets

A fab should not be defined only by wafer starts. It should be defined by the products it intends to manufacture.

### Target Product Portfolio

| Product Type        | Typical Process Node |
|---------------------|----------------------|
| Automotive MCU      | 40nm - 90nm          |
| Industrial MCU      | 40nm - 130nm         |
| Connectivity IC     | 28nm - 65nm          |
| Display Driver IC   | 28nm - 55nm          |
| Network Processor   | 28nm - 45nm          |
| Edge AI Controller  | 28nm                 |
| Industrial SoC      | 28nm - 45nm          |
| Security Controller | 40nm - 90nm          |

### Typical Die Characteristics

| Metric                   | Assumption             |
|--------------------------|------------------------|
| Average Die Size         | 50-150 mm <sup>2</sup> |
| Gross Dies / 300mm Wafer | ~400-1300              |
| Functional Yield Target  | 85-95%                 |
| Wafer Sort Yield         | 95-99%                 |
| Packaging Yield          | 98-99.5%               |
| Final Test Yield         | 95-99%                 |

*These values are illustrative and product dependent.*

A 50mm<sup>2</sup> die and a 150mm<sup>2</sup> die processed on the same node can have dramatically different economics because gross die count and defect sensitivity change with die area.

These assumptions will later drive:

- Revenue
- Material consumption
- Yield
- Cost per good die
- Packaging requirements
- Customer economics

## 2.3 Required site elements

A 40,000 WSPM 300mm fab requires more than a cleanroom. The manufacturer must plan for:

- Land
- Fab building
- Cleanroom
- Sub-fab
- Office and support buildings
- Utility buildings
- Chemical storage and distribution
- Specialty gas farms
- Bulk gas systems
- Wastewater treatment
- Electrical substation
- UPW plant
- Chillers
- HVAC systems
- Exhaust and abatement
- Warehouse
- Logistics docks
- Maintenance shops
- Training center
- Emergency response infrastructure
- EHS infrastructure
- Security infrastructure
- Data and manufacturing-control infrastructure

## Representative FAB value-creation process flow:



Figure 2: Semiconductor fab value-creation flow, illustrating the progression from site and infrastructure readiness to qualified products, customers, and cash generation.

FULL VERSION AVAILABLE ON REQUEST

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